Hybrid Power Optimization at the Libyan Investment Authority: A synergy of Homer Software and Machine Learning application

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***Abstract*— There is a global demand for clean, safe, reliable, and stable electrical energy. This paper presents an optimal sizing of a photovoltaic generator (PVG), Wind turbines (WT), and an energy storage sources (ESS) integrated in a company using Homer software and Machine learning. The case study focuses on the Libyan Investment Authority, which is one of the sovereign and investment funds with significant financial investment assets worldwide the significant work is to find a solution to the problem of electricity supply to the Authority, which is considered completely unacceptable due to the sensitivity of the sovereign fund of the Libyan state. The objective of the study is to cover 250 kW of power demand in the company by integrating renewable energy sources, ESS, and diesel generator. Consequently, the study examines the electric, financial, and environmental aspects of the project. The total system production amounts to 2,398,024 kWh per year, resulting in a total cost of $5.69 million.**

***Keywords*— optimal sizing, renewable energy sources, energy storage sources, Homer software, Machine learning.**

I. INTRODUCTION

There is an undeniable global shift occurring in the field of electricity demand. This rapid expansion is particularly evident in emerging countries that possess significant economic potential [1]. Therefore, it is crucial to integrate renewable energy sources (RES) to cover the needs of isolated villages, ensuring their self-sufficiency through the exploration of additional energy sources. Wind energy and solar energy are the two most significant alternative energy sources due to their abundant availability, making them viable options to address the electricity demands of these communities [2].

Numerous studies have been conducted to this field. In [3], the design of solar energy systems was carried out based on the designs of PVSYST software, where all the generated results were dependent on the amount of energy demanded from buildings and villages. Another research in [4], has focused on integrating wind turbines along with Li-ion batteries, which are considered as energy storage system (ESS). The researchers considered the degradation of battery performance, its lifespan, its high cost, and the fluctuation of the yield. Therefore, they resorted to use battery ESS to address these challenges. Battery ESS helps reduce energy fluctuations, and an algorithm based on Dynamic Programming was utilized for calculating costs and power. Another study in [5] has conducted a comparison of results from multiple algorithms. The Whale Optimization Algorithm, Slap Swarm Algorithm, Water Cycle Algorithm, and Grey Wolf Optimization were implemented using the MATLAB software. Subsequently, the various results were compared and evaluated, and the most suitable one was selected. In certain energy storage studies [6], the system was integrated into a hybrid system using the Generalized Reduced Gradient algorithm. In the field of optimal cost offerings [7], researchers concentrated on designing clean electricity supply and production systems by simulating the Homer Software to arrive at the most suitable design proposal with the lowest cost. In certain studies [8], the Battery Energy Storage System (BESS) was employed to shift the cycle period of electric power production from daytime to midnight. The costs and market conditions were considered to further optimize the battery capacity when it receives power from various generation systems. Nevertheless, each of the these studied is interested in only one or two particular objectives (electrical, economical, and environmental). This paper presents an optimal sizing of RES, ESS, and diesel generators using Homer software.

The case study focuses on the Libyan Investment Authority. The main objective is to cover 250 kW of its power demand by integrating RES, ESS, and diesel generators and using Homer software and Machine learning. In the result, three main axes are presented to prove the efficiency of the study: electrical, economical, and environmental. This paper is structured as fallows: section 2 introduces the studied system; Section 3 provides a description of the framework design. And the results of the case study company are presented in the final section.

II. STUDIED SYSTEM

In this work, a hybrid clean energy sources are joined to a company as clarify Fig. 1. Overall, companies play an essential part of our society and economy by distributing valuable services and needs, providing jobs, driving innovation, and taking environmental and social responsibilities. To enhance the company productivity and electricity stability, the studied company choose to incorporate sustainable development practices into their business models by planting RES. It joins hybrid clean energy sources that consist of a PVG and WT. Yet, they are two intermittent sources as they depend on weather conditions. For this reason, an ESS is added as it ensures that RES might be used consistently to facilitate the widespread use of solar and wind energy. In addition, the system integrates diesel generators that act as a backup power source during power outages or disruptions.

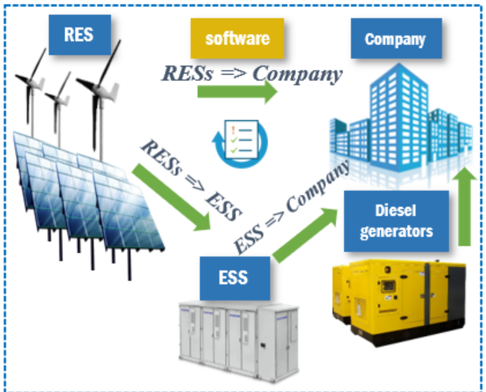

Fig. 1. Studied company hybrid energy sources design

III. Framework Design Description

A. Homer Software

This software, developed by the National Renewable Energy Laboratory of the United States, allows for the design of micropower systems, whether connected to the grid or not [9]. It offers a comprehensive computer model simulation program that enables the study of various technologies, including energy production and storage, based on the required load. The program considers the economic aspects of the design in terms of the cost of construction and the cost of maintenance and replacement for different periods of time. Five important steps can summarize the general working process of the homer software, as in Fig. 2 [10]:

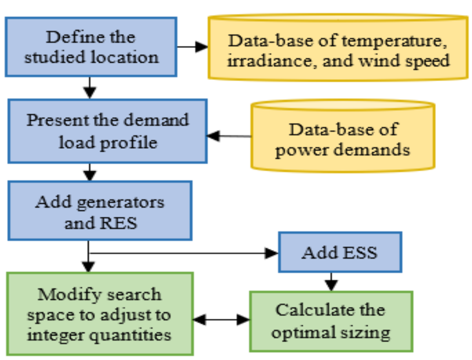

Fig. 2. Homer software essential steps

- Defining the location in order to obtain the yearly irradiance, temperature, and wind speed profiles.
- Presenting the studied company load profile.
- Adding power grid and RES generators as energy resources.

In this step, essential information such as the constraints of maximum/minimum power generation, capital, and maintenance costs are designed.

- Adding ESS with all economical and technical information.
- Calculating the optimal sizing of the chosen components.

B. Sizing methodology

The sizing of the studied devices; PVG, WT, ESS, and diesel generator is presented in Fig. 3:

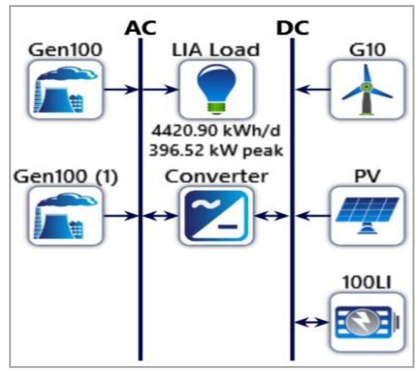

Fig. 3. Proposed hybrid microgrid structure for LIA

The optimization process within HOMER is a dynamic and iterative approach that takes into consideration a multitude of essential factors:

System Reliability: HOMER prioritizes a dependable energy supply by evaluating component interactions to minimize the risk of energy shortages. It also integrates redundancy and fail-safes to enhance system reliability.

Cost-Efficiency: HOMER meticulously evaluates and minimizes costs associated with component acquisition, operation, and maintenance to achieve the most cost-effective configuration. This typically involves prioritizing renewable sources like PV and wind over more expensive options such as diesel generators.

Environmental Sustainability: HOMER assesses the system's environmental impact, aiming to minimize carbon emissions, reduce pollution, and promote a cleaner, more sustainable energy environment.

Hybrid microgrid production: HOMER streamlines the design and operation of hybrid microgrids by considering various energy sources production.

- PVG generated power

The generated PVG power $\boldsymbol{P_{PVG}}$ is calculated using Equation (1).

$$\boldsymbol{P_{PVG} = Y_{PVG} f_{PVG}\left(\frac{\overline{G}}{\overline{G}_{STC}}\right)[1 + \alpha_1(T - T_{STC})} \quad (1)$$

Where:

• $\boldsymbol{Y_{PVG}}$: the power PVG rated output with standard test conditions ($\boldsymbol{STC = 25°C}$),

• $\boldsymbol{f_{PVG}}$: the PVG derating factor.

•$\boldsymbol{\overline{G}}$: the currently PVG solar irradiation.

• $\overline{G_{STC}}$: the incident irradiation at STC.
• α_1 : the temperature coefficient of power.
• T: the currently PVG cell temperature.
• T_{STC}: the PVG cell temperature with the STC.

- WT generated power

The equation of the WT power output is expressed as follows.

$$P_{WT} = \left(\frac{\rho}{\rho_0}\right) . P_{WT,STC} \qquad (2)$$

Where:
• P_{WT}: the WT power output.
• ρ: the current air density.
•ρ_0:the air density at STC ($\rho_0 = 1.225$ kg/m^3).
•$P_{WT,STC}$: the WT power output at STC.

- Energy Storage System

The mathematics expressions of the battery charge model, its maximum charge rate, and its maximum charge current values, that are calculated by HOMER software, are given in Equations 4-6 [11]. The battery charge power is expressed in (7) and the battery discharge power is expressed in (8).

$$P_{batt,max} = \frac{KQ_1 e^{-k\Delta t} + Qkc(1 - Qe^{-k\Delta t})}{1 - e^{-k\Delta t} + c(k\Delta t - 1 + e^{-k\Delta t})} \qquad (3)$$

$$P_{batt,str,max} = \frac{(1 - e^{-\alpha_2 \Delta t})(Q_{max} - Q)}{\Delta t} \qquad (4)$$

$$P_{batt} = \frac{N_{batt} I_{max} V_{nom}}{1000} \qquad (5)$$

$$P_{batt,ch} = \frac{MIN(P_{batt,abs,max}, P_{batt,c\,str,max}, P_{batt,c\,str,max})}{\eta_{batt,c}} \qquad (6)$$

$$P_{batt,dis} = \eta_{batt,d} \frac{-KcQ_{max} + kQ_1 e^{-k\Delta t} + Qkc(1 - Qe^{-k\Delta t})}{1 - e^{-k\Delta t} + c(k\Delta t - 1 + e^{-k\Delta t})} \qquad (7)$$

Where:

• Q_1: the amount of the available energy in the storage at the beginning.
• Q: the total amount of energy in the storage at the beginning.
• C: the storage capacity ratio.
• k: the storage rate constant.
•Δt: the time period.
•α_2: the storage maximum charge rate.
• Q_{max}: the total capacity of the ESS.
• N_{batt}: the number of batteries in the ESS
• I_{max}: the storage maximum charge current.
• V_{nom}: the storage nominal voltage.
•η_{batt}: the storage charge efficiency.

- Diesel generator

During electric energy interruption, diesel generator is used. to provide a robust energy supply. The diesel-fueled of the diesel generator expression is presented as fallow [11].

$$F = F_x Y_{gen} + F_y P_{gen} \qquad (8)$$

where:
• F: the hourly fuel consumption.
• F_x: the coefficient of the fuel curve intercept.
• F_y: the slope of the fuel curve.
• Y_{gen}: the generator rated capacity.
• P_{gen}: the generator electrical output.

C. *Machine Learning*

Machine learning can be described as the capability of computer systems to generate accurate predictions by leveraging past experiences. Four fundamental steps can be challenging to explain its prediction: data collection, data processing, input data split into train data and test data, and machine learning techniques selection:

- Random Forest: it utilizes bagging techniques. In a Random Forest, each tree makes its individual prediction, and the model's ultimate decision is determined by the majority of votes from these trees, as illustrated in Fig. 4 [13].

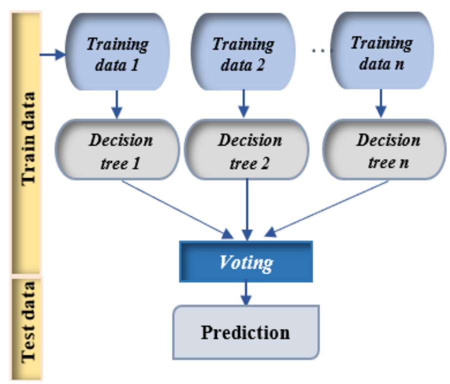

Fig. 4. Random Forest technique.

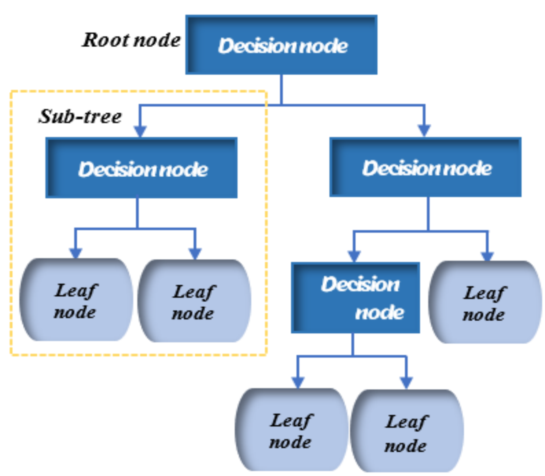

Fig. 5. Decision tree technique.

- Decision Tree: it starts with the root node that partitions the entire dataset into distinct homogeneous sets based on specific conditions associated with dataset features. This process continues until a leaf node is reached, at which point the output class associated with that leaf node is disclosed, as illustrated in Fig. 5 [14].

IV. Result and discussion: case study

A. *Libyan Investment Authority*

The Libyan Investment Authority, which its location is presented in Fig. 6, is a government holding company that is considered as a sovereign wealth fund for the Libyan state. It was established on August 28, 2006, according to Resolution No. 208 issued by the Libyan government at the time.

Fig.6. Studied location

The Corporation manages several investments in various fields, including agriculture, real estate, finance, oil, and gas. It also manages several subsidiaries. Therefore, the Corporation has extremely important digital data and extremely sensitive electrical requirements that need to ensure the provision of energy away from frequent network outages and out of any dangers that occur in a crisis.

The company power demand devices, are summarized in Table I.

TABLE I. Libyan Investment Authority power demand

Company devices	**Power (kW)**
Server room 121kw	122 kW
air conditioners	26 kW
Administration Building	20 kW
Water wells and pump	52 kW
Lighting	30 kW
Total Load Distribution	250 kW

From the table cited above, it can be considered that the company can reach a maximum energy consumption equal to 250kW. Additionally, the most significant portion of the company's power consumption is allocated to the server room. Figure 7 shows the yearly entire company power consumption and the yearly server room consumption.

B. *Machine learning techniques scores*

Including the server room's power consumption as a feature when predicting the company's total power consumption proves presents a valuable strategy. This inclusion enables the use of pertinent data and the potential identification of causal relationships or correlations.

As a result, it facilitates precise predictions, empowering data-driven decision-making and the ongoing enhancement of the model's performance. Moreover, this approach improves the model's interpretability and assists in optimizing energy utilization for the company's advantage.

To construct the optimal model for the provided dataset, two methods are employed: Random Forest and Decision Tree. The main objective of using ML is to facilitate to predict the company whole consumption and the its energy management in other works.

Table II shows the scores of both techniques after training and testing. As indicated, the second technique accurately produces the correct test outputs, unlike the first one.

TABLE II. techniques scores.

Technique name	*Score*
Random forest	92.2 %
Decision Tree	100.0%

C. *Hybrid system Homer software components*

After starting the simulation using Homer software, 12 different combinations were obtained. The results gave us the required power of the PVG, WT, diesel generator, and system converter and the ESS capacity as an optimal combination with a total production equal to 236.9kWh. In the studied microgrid, supplying diesel generators and ESS is crucial in case of emergency.

After selecting the optimal category, the Homer software displays the sizing of each component as given in Table III.

TABLE III. Studied system sizing

Component	**Name**	**Size**
Diesel generator	Generic 100kw fixed capacity genset	100*2 kW
PVG	Generic flat plat PV	555 kW
WT	Generic 10 kW	119*7 kW
ESS	Generic 100kWh Li-Ion	43 strings
System converter	System Converter	289 kW
Dispatch strategy	HOMER cycle charging	

For the sake of providing the electric service continuity for the Libyan Investment Authority, 86.1% of total energy is produced by RES. All this production is directly furnished to the load. According to Table IV, the amount of the PVG generated power is equal to 33.9%. The remaining 13.9% of power is generated from the diesel generators.

TABLE IV. Studied system energy production.

Component	Production (kWh/Year)	%
Diesel generator 1	236,900	9.86
Diesel generator 2	96,910	4.04
PVG	812,242	33.9
WT	1.251,972	52.2
Total	2,398,024	100

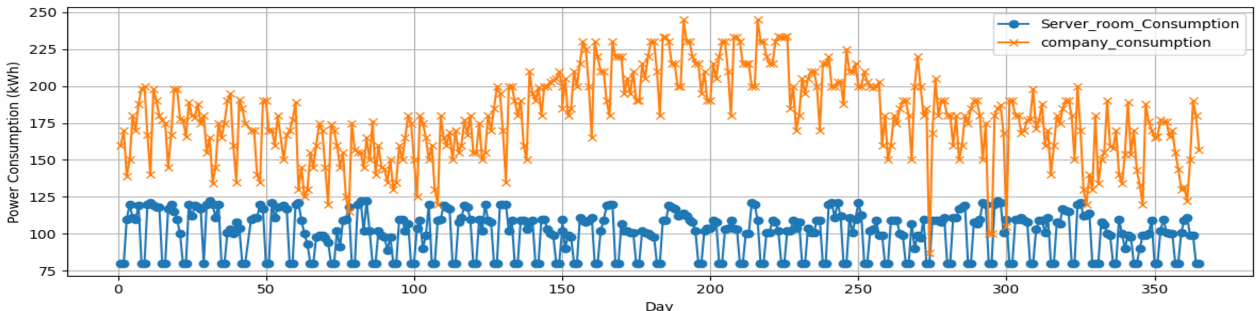

Fig. 7. Power consumption

the diesel generators statistics are presented in Table V. It includes hours of operations, operational life, fixed and marginal generation costs. It is important to note that the operational life is an estimate and not an absolute guarantee, it depends on many factors, such as, the maintenance and the operation conditions. Also, it is important to clarify the difference between the fixed and the marginal generation costs:

- Fixed generation cost known as the capital cost, it presents the costs accrued during the establishment of the power generation infrastructure, as, construction, equipment, and initial investment.
- marginal generation cost known as the variable cost, it presents the extra costs generated for producing each additional unit of electricity, as, fuel cost or diesel [15].

TABLE. V. Diesel generators statistics

Quantity	Value (Generator 1 - Generator 2)
Hours of operation	2,841 - 993 hrs/yr
Operational life	5.28 - 15.1 yr
Fixed generation cost	7.47 $/hr
Marginal generation cost	0.253 $/kWh

Table VI illustrates some characteristics of the chosen PVG and WT, the maximum output, and the rated capacity. The table includes the hours of operation, the levelized cost. In fact, the main role of this cost is the evaluation of a project or investment total cost over its lifetime [16]. It considers many factors as the operating cost and the maintenance expenses. For that, it is considered as a valuable decision-making tool within the energy industry, as it enables the assessment of cost-effectiveness and competitiveness among different energy sources.

TABLE VI. PVG and WT electric summary

Quantity	PVG	WT	Unit
Maximum output	530	1,190	kW
Rated Capacity	555	1,190	kW
Hours of operation	4,346	6,469	hrs/yr
Levelized Cost	0.139	0.0666	$/kWh

Table VII presents some result data of the chosen ESS. It includes the average energy cost, the stored and injected energy, and the storage depletion. As fact, the average energy cost, refers to the expenditure associated with each unit of the stored or injected energy throughout the battery operational life. Also, it depends on many factors such as the battery size and capacity, the charging/discharging rates, and the electricity cost.

TABLE VII. ESS result data (1)

Quantity	Value	Unit
Average energy cost	0.0447	$/kWh
Energy In	629,951	kWh/yr
Energy Out	568,570	kWh/yr
Storage depletion	1,702	kWh/yr

In addition, Table VIII includes the nominal capacity of the battery, its autonomy, and its expected life. For clarify, the battery autonomy refers to the period of time that a battery can function or supply power without the need for recharging or external energy sources. The table indicates that the selected battery autonomy is 18.7 hours. Since the system requires the integration of three batteries, the total autonomy is 56.1 hours. This autonomy can be considered highly favorable in terms of meeting the power needs of the system.

TABLE VIII. ESS result data (2)

Quantity	Value	Unit
Nominal Capacity	4,300	kWh
Autonomy	18.7	Hr
Expected Life	15.0	Yr

The system converter presents a crucial interface between the diesel generator and the WT, PG, and ESS. It enables efficient energy conversion. Tables IX and X, illustrate the summary and statistics of the chosen system converter in the studied case. The lowering of the energy output compared to the energy input can be attributed to power losses due to many factors as transformers losses, and resistance in conductors.

TABLE IX. System converter summary

Quantity	Value	Unit
Hours of operation	8,025	hrs/yr
Energy In	1,371,923	kWh/yr
Energy Out	1,303,327	kWh/yr

The assessment of the hybrid system in terms of costs is presented in Figure 8. It includes, the capital, operating, replacement, salvage, resource, and the total cost. The highest capital in the system pertains to the PVG system and the highest operating cost pertains to the ESS. The total cost of the entire system is predicted as $5.69M.

TABLE X. System converter statistics

Quantity	Value	Unit
Maximum Output	289	kWh/yr
Capacity Factor	51.1	%

In the studied case, the analysis of the environmental dimension is essential, as the system includes two diesel generators. For that, integrating renewable energy as a green technology to meet 250 kW of the company's energy demand can reduce the quantity of greenhouse gas emissions. Tables XI and XII present the diesel consumptions statistics and emission.

TABLE XI. Diesel Consumption Statistics

Quantity	Value	Unit
Total fuel consumed	95,189	L
Avg fuel per day	261	L/day
Avg fuel per hour	10.9	L/hour

TABLE XII. Emission

Quantity	Value	Unit
Carbon Dioxide	248,975	kg/yr
Carbon Monoxide	1,694	kg/yr
Unburned Hydrocarbons	68.5	kg/yr
Particulate Matter	6.78	kg/yr
Particulate Matter	610	kg/yr
Nitrogen Oxides	136	kg/yr

V. CONCLUSION

The objective of this study is to generate clean and reliable energy for the Libyan Investment Authority, ensuring electricity provision while preserving the environment and the economy. The purpose was covering 250 kW of power demand in the company with combining wind and solar energy, with Li-ion batteries for energy storage. In emergency situations, diesel generators are utilized. The total energy production amounts to 2,398,024 kWh/Year, resulting in a total cost of $5.69 million. Additionally, the study found that the ESS plays a significant role in reducing the reliance on diesel generators, thereby minimizing gas and carbon emission.

Hence, by embracing sustainable development, companies not only make a positive impact on environmental and social advancement, but also bolster their standing, appeal to mindful consumers, manage risks effectively, and secure a competitive edge in the long term.

References

[1] M M Rahman. “Environmental degradation: The role of electricity consumption, economic growth and globalisation”. J. Environ. Manage. Vol. 235, 2020, 109742.

[2] M Ben Arab, M Rekik, L Krichen. “A priority-based seven-layer strategy for energy management cooperation in a smart city integrated green technology”. Appl. Energy. Vol. 335, 2023, 120767.

[3] M Çöl, Ö Köylüoğlu. “Renewable Energy Sources Energy Policy and Energy Management Capacity Calculation and Subsidization Proposals for Rooftop PV Energy for a Residential Building in İstanbul”. Renewable Energy Sources Energy Policy and Energy Management. Vol. 2, 2021, pp. 1-23.

[4] Y Liu, X Wu, J Du, Z Song, G Wu. “Optimal sizing of a wind-energy storage system considering battery life”. Renew. Energy. Vol. 147, pp. 2470-2483.

[5] A Diab, H M. Sultan, O Kuznetsov. “Optimal sizing of hybrid solar/wind/hydroelectric pumped storage energy system in Egypt based on different meta-heuristic techniques”. Environ. Sci. Pollut. Res. Vol. 27, 2020, pp. 32318–32340.

[6] L Al-Ghussain, A D Ahmad, A M. Abubaker, M A. Mohamed. “An integrated photovoltaic/wind/biomass and hybrid energy storage systems towards 100% renewable energy microgrids in university campuses”. Sustain. Energy Technol. Assess. Vol. 46, 2021, pp. 101273.

[7] P Pal, V. Mukherjee. “Off-grid solar photovoltaic/hydrogen fuel cell system for renewable energy generation: An investigation based on techno-economic feasibility assessment for the application of end-user load demand in North-East India”. Sustain. Energy Technol. Assess. Vol. 149, 2021, pp. 111421.

[8] Y Yoo, G Jang, S Jung. “A Study on Sizing of Substation for PV With Optimized Operation of BESS”. Special section on evolving technologies in energy storage systems for energy systems applications.
Vol. 8, 2020.

[9] O Ekren, C H Canbaz, Ç B Güvel. “Sizing of a solar-wind hybrid electric vehicle charging station by using HOMER software”. J. Clean. Prod. Vol. 279, 2021, 123615.

[10] J B Jiménez, E F Caicedo Bravo. “Optimal sizing of a grid-connected microgrid and operation validation using HOMER Pro and DIgSILENT”. Scientia et Technica Año. Vol. 27, 2022, pp. 2344-7214.

[11] İ Çetinbaşa, B Tamyüreka, M Demirtaş. “Design, Analysis and Optimization of a Hybrid Microgrid System Using HOMER Software: Eskişehir Osmangazi University Example”. Int. J. Renew. Energy Dev. Vol. 8, 2019, pp. 65-79.

[12] R. Venkateswari, S. Sreejith. “Factors influencing the efficiency of photovoltaic system”. Renew. Sustainable Energy Rev. Vol. 101, 2019, pp. 376-394.

[13] M. Sušac, A. Has, M. Knezˇevic “Predicting energy cost of public buildings by artificial neural networks, CART, and random forest” Neurocomputing. Vol. 439, 2021, 223-233.

[14] M. Bansal, A. Goyal, A. Choudhary “A comparative analysis of K-Nearest Neighbor, Genetic, Support Vector Machine, Decision Tree, and Long Short Term Memory algorithms in machine learning”. Decis. Anal. Vol. 3, 2022, pp. 100071.

[15] M. Bhoopathi1, P. Palanivel, “Estimation of Locational Marginal Pricing Using Hybrid Optimization Algorithms”. Intell. Autom. Soft Comput. Vol. 31, 2022, 1.

[16] W Shen, X Chen, J Qiu, G Hayward, S Sayeef, P Osman, K Meng, Z Y Dong. “A comprehensive review of variable renewable energy levelized cost of electricity”. Renew. Sustainable Energy Rev. Vol. 133, 2020, 110301.